

## Project Workflow: Week 2

**Project:** EasyStego: A Drag-and-Drop Image Steganography Tool

**Phase:** Requirements Analysis & System Design

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**Course:** Software Development I (CSE 2216)

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### 1. Objective

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The primary goal of Week 2 was to complete requirements analysis, design the system architecture, and produce the initial project documentation. This included:

- Gathering all functional and non-functional requirements for the EasyStego application.
- Designing the system workflow for both the Hide Message and Reveal Message operations.
- Creating basic wireframes for the two-tab interface and drag-and-drop zones.
- Writing the initial README.md documentation to outline the project purpose, features, and usage instructions.

### 2. Detailed Task Breakdown

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**Requirements Gathering:** Identified all core functional requirements — LSB-based text embedding, message extraction, drag-and-drop image loading, real-time capacity calculation, one-click clipboard copy, and error dialogs for unsupported formats and oversized messages.

**System Workflow Design:** Mapped out the two primary data flows: (1) the Embedding Process — image input → validate → user message input → append `####END####` terminator → LSB write → save PNG; and (2) the Extraction Process — stego-image input → read pixel LSBs → reconstruct bytes → detect terminator → display message.

**UI Wireframing:** Sketched the layout of the two-tab interface (Hide Message / Reveal Message). Each tab contains a dashed-border drop zone at the top, a text area in the center, a capacity/status label, and an action button at the bottom. Image previews are rendered inline within the drop zone after a file is loaded.

**README.md Documentation:** Drafted the project README.md covering: project overview and features, how the LSB algorithm works (embedding and extraction), the capacity formula ( $\text{width} \times \text{height} \div 8$ ), step-by-step usage instructions for both tabs, the tech stack table (Java, Swing, ImageIO, VS Code, Git), project file structure, known limitations, and future work.

**Non-Functional Requirements:** Defined constraints including cross-platform compatibility (Windows, macOS, Linux), PNG-only output to preserve embedded bits, JDK 11+ as the minimum runtime, and visually imperceptible output images.

### 3. Deliverables

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- Requirements Document: a complete list of functional and non-functional requirements for EasyStego.
- System Workflow Diagram: a step-by-step flow for both the embedding and extraction processes.
- UI Wireframe Sketches: layout diagrams for the Hide Message and Reveal Message tabs.
- README.md: initial project documentation covering features, usage, algorithm explanation, tech stack, and limitations.

## 4. Tools & Resources

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### Development Environment:

- Java JDK 11+
- Visual Studio Code
- Git / GitHub — for version control and README hosting

### Reference Materials:

- Java Swing documentation (javax.swing, java.awt.datatransfer)
- javax.imageio.ImageIO — for PNG read/write operations
- Markdown syntax guide — for README.md formatting